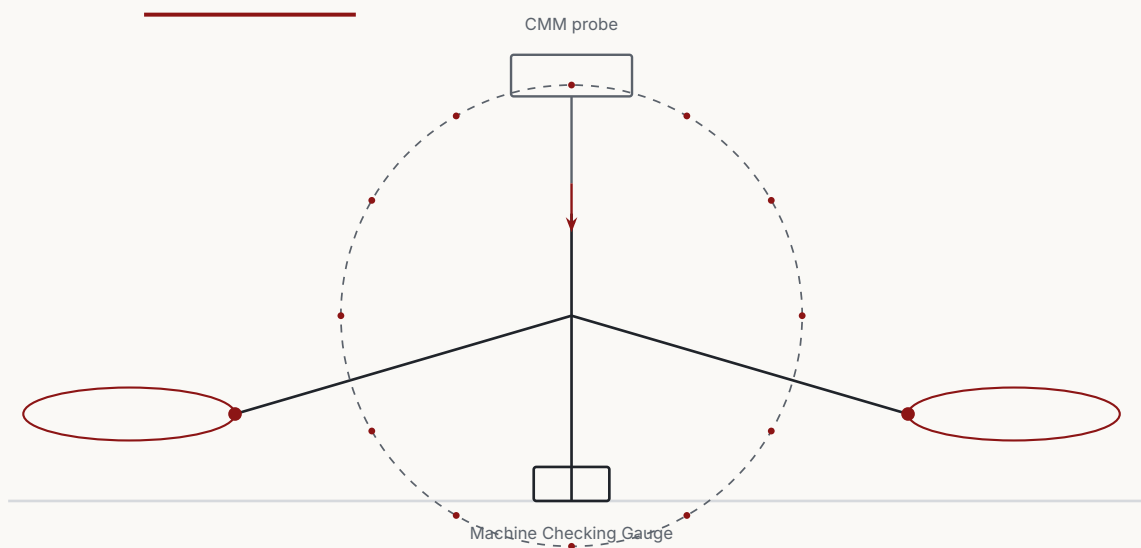


DIMENSIONAL METROLOGY

Spatial Verification of a Coordinate Measuring Machine

Renishaw Machine Checking Gauge trajectory design,
geometric-error identification, hysteresis assessment,
and extended workspace coverage



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Abstract

This report develops a metrologically rigorous workflow for periodic verification of a Cartesian coordinate measuring machine (CMM) using a Renishaw Machine Checking Gauge (MCG). The project combines numerical trajectory generation, ZEISS CMM execution, planar circularity assessment, three-dimensional error identification, and a strategy for extending the spatial coverage of the verification volume. Nominal points and unit probing normals are generated for circular and spherical trajectories, exported to the CMM software, and executed with both unidirectional and reversal-containing scan sequences in order to expose direction-dependent behavior.

The planar experiments report circularity deviations between $1.10\text{ }\mu\text{m}$ and $1.49\text{ }\mu\text{m}$. The two unidirectional two-revolution paths average $1.115\text{ }\mu\text{m}$, approximately 22% below the $1.43\text{ }\mu\text{m}$ average of the reversal-containing paths. This pattern is consistent with a direction-history effect, although the available data do not by themselves separate mechanical hysteresis from thermal drift, probing repeatability, qualification error, or numerical association effects. For the spatial study, normal-direction deviations are modeled by a linear least-squares problem that estimates center offsets, effective arm-length error, and small axis-squareness terms. The translation and radius components of the synthetic validation are recovered accurately, while two angular channels show significant scaling inconsistency; the report therefore recommends a joint singular-value-decomposition solution and explicit conditioning analysis before the angular estimates are used for compensation.

A revised verification protocol is proposed using equal-area spherical sampling, continuous one-direction reference paths, paired reversal tests, repeated runs, thermal monitoring, uncertainty budgeting, and complementary upper and lower spherical segments. The resulting method distinguishes diagnostic filtering from traceable measurement results and aligns the project with the general principles of ISO 10360 verification and the GUM framework for measurement uncertainty.

KEYWORDS coordinate measuring machine; Machine Checking Gauge; circularity; sphericity; least squares; axis squareness; hysteresis; measurement uncertainty; spatial coverage.

Executive Summary



Renishaw Machine Checking Gauge used as a large-radius kinematic artefact for CMM verification.

ENGINEERING OBJECTIVE

Develop a repeatable protocol that expands the spatial verification coverage of a ZEISS CMM while identifying direction-dependent and geometric errors.

TRAJECTORY DESIGN

Circular and spherical paths generated in Python for six nominal arm lengths; the reported study uses a 151 mm arm, 60 points per planar revolution, and ten latitude rings over $\pm 45^\circ$.

PLANAR RESULT

Reported circularity spans 1.10–1.49 μm ; paths without reversal produce lower average deviation than reversal-containing paths.

SPATIAL MODEL

Linear least-squares estimation of translation, effective radius, and small squareness terms from the scalar normal deviations returned by the CMM.

RECOMMENDED PROTOCOL

Equal-area ring sampling, unidirectional baselines, matched reversal pairs, repeated runs, SVD-based estimation, explicit uncertainty budgets, and complementary spherical segments.

The experimental figures, original equations, and reported numerical values are retained from the submitted project. The English prose, report architecture, theoretical development, statistical interpretation, and metrological cautions have been expanded for publication-quality presentation. Internal inconsistencies are identified explicitly rather than silently corrected.

Nomenclature

CMM	Coordinate measuring machine.
MCG	Machine Checking Gauge.
$\mathbf{p}_i$	Nominal Cartesian coordinates of measurement point i .
$\mathbf{n}_i$	Unit probing normal associated with point i .
ε_i	Scalar deviation returned by the CMM along $\mathbf{n}_i$.
$\mathbf{q}_i$	Reconstructed measured point, $\mathbf{q}_i = \mathbf{p}_i + \varepsilon_i \mathbf{n}_i$.
$R, \delta R$	Nominal MCG radius and effective radius error.
u, v, w	Estimated translational offsets of the realized sphere center.
α, β, γ	Small axis-squareness parameters under the project sign convention.
N, N_j	Number of points on a circle and on latitude ring j .
θ, ϕ	Azimuth and elevation angles.
t_{CIR}	Minimum-zone circularity deviation.
t_{SPH}	Minimum-zone sphericity deviation.
$H_{\text{RMS}}, H_{\text{pp}}$	RMS and peak-to-peak direction-reversal hysteresis indicators.
u_c, U	Combined standard uncertainty and expanded uncertainty.
$\kappa(A)$	Condition number of the least-squares design matrix.

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Introduction

Coordinate measuring machines are used to establish the geometry of manufactured parts by combining calibrated axis scales, a probing system, numerical feature association, and environmental compensation. Their reported coordinate values are therefore the output of an entire measurement chain rather than the direct reading of a single sensor. Periodic verification is required to detect changes in volumetric accuracy, probing behavior, axis squareness, structural stability, and thermal response before those changes affect product conformity decisions.

The Machine Checking Gauge provides a large-radius kinematic artefact whose arm rotates about a mechanical center. When the probe follows the moving stylus location, the nominal trajectory is a circle in a plane or a spherical segment in space. Because the nominal radius is much larger than that of a conventional reference sphere, the artefact can interrogate a substantial portion of the CMM working volume and can expose direction-dependent effects that are not visible in a local probe qualification test.

The original project had two objectives. The first was to implement circular and spherical MCG trajectories on a ZEISS CMM, compare theoretical points with measured deviations, and investigate the apparent sensitivity near $\pm 45^\circ$. The second was to propose a practical strategy for improving coverage of the useful machine volume, notably by combining complementary spherical segments. This research edition retains that experimental work and strengthens it through a formal measurement model, explicit form-error definitions, least-squares conditioning, hysteresis metrics, thermal and uncertainty analysis, and a structured verification protocol.

The report is organized as follows. [Chapter 2](#) establishes the mathematical and metrological framework. [Chapter 3](#) describes the generation and execution of planar and spatial trajectories. [Chapter 4](#) evaluates the four planar result sets. [Chapter 5](#) develops the spatial identification model and audits its numerical validation. [Chapter 6](#) proposes an improved coverage and verification strategy. The final chapter summarizes the findings and defines the actions required before the method can support traceable acceptance decisions.

Scope and interpretation

The project is a diagnostic and verification study, not a complete ISO 10360 acceptance test. A form plot with a low numerical deviation is encouraging, but conformity requires a declared test procedure, calibrated artefacts, environmental records, decision rules, and an uncertainty statement.

Metrological Foundations and Error Model

WORK PACKAGE 1 Measurement chain, geometric association, and uncertainty

2.1 CMM verification context

The ISO 10360 series provides the principal framework for acceptance and reverification tests of coordinate measuring systems. ISO 10360-2 addresses CMMs used for linear dimensions, whereas ISO 10360-5 addresses contacting probing systems. The present MCG campaign complements those standardized tests by providing a large-radius, path-dependent diagnostic of the machine volume. The results should therefore be interpreted as evidence of machine behavior and as input to a maintenance protocol, not as a replacement for the standardized verification quantities.[1, 2]

A complete measurement result may be represented as

$$Y = f(X_1, X_2, \dots, X_m),$$

where the input quantities include axis scale readings, probe qualification parameters, artefact geometry, temperature, numerical fit parameters, and repeatability effects. In the GUM framework, the combined standard uncertainty is obtained from the sensitivity coefficients $c_j = \partial f / \partial X_j$:

$$u_c^2(Y) = \sum_{j=1}^m c_j^2 u^2(X_j) + 2 \sum_{j < k} c_j c_k u(X_j, X_k).$$

When correlations are negligible, the covariance terms vanish. An expanded uncertainty is then reported as

$$U = k u_c,$$

with $k \approx 2$ commonly used for an interval close to 95% coverage when the effective degrees of freedom are sufficiently large.[5]

2.2 Nominal point and normal-direction deviation

For each commanded point, the ZEISS program receives a nominal coordinate vector

$$\mathbf{p}_i = \begin{bmatrix} x_i & y_i & z_i \end{bmatrix}^T$$

and a unit probing direction

$$\mathbf{n}_i = \begin{bmatrix} n_{x,i} & n_{y,i} & n_{z,i} \end{bmatrix}^T, \quad \|\mathbf{n}_i\|_2 = 1.$$

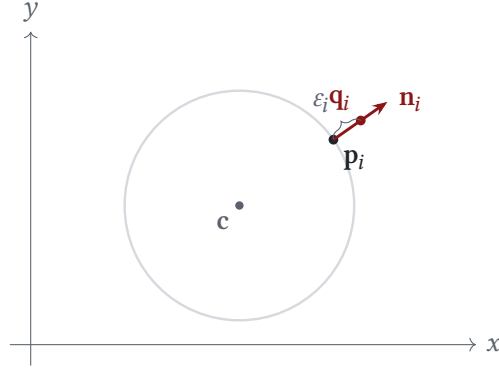


Figure 2.1: Normal-direction measurement model used to reconstruct each measured point.

The CMM returns a scalar deviation ε_i along the probing normal. A physically interpretable point can therefore be reconstructed as

$$\mathbf{q}_i = \mathbf{p}_i + \varepsilon_i \mathbf{n}_i.$$

This expression is the central link between the theoretical trajectory and the experimental data. It also makes clear that a scalar normal residual cannot reveal tangential error components unless the path geometry and normal directions vary sufficiently throughout the campaign.

2.3 Parametric circle and sphere models

For a circle of radius R in the XY plane, centered at $\mathbf{c} = (c_x, c_y, c_z)^\top$, the nominal points and outward normals are

$$\mathbf{p}(\theta) = \mathbf{c} + R \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix}, \quad \mathbf{n}(\theta) = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix}.$$

With N evenly spaced points,

$$\theta_i = \theta_0 + \frac{2\pi i}{N}, \quad i = 0, \dots, N-1.$$

For a spherical trajectory parameterized by azimuth θ and elevation ϕ ,

$$\mathbf{p}(\phi, \theta) = \mathbf{c} + R \begin{bmatrix} \cos \phi \cos \theta \\ \cos \phi \sin \theta \\ \sin \phi \end{bmatrix}, \quad \mathbf{n}(\phi, \theta) = \frac{\mathbf{p}(\phi, \theta) - \mathbf{c}}{R}.$$

The original experiment restricts the elevation to

$$-45^\circ \leq \phi \leq 45^\circ,$$

so that

$$-R \sin 45^\circ \leq z - c_z \leq R \sin 45^\circ$$

and the radius of each latitude ring is

$$\rho(\phi) = R \cos \phi \in [R \cos 45^\circ, R].$$

2.4 Associated features and form error

The least-squares associated circle is obtained from

$$(\hat{\mathbf{c}}, \hat{R}) = \arg \min_{\mathbf{c}, R} \sum_{i=1}^N (\|\mathbf{q}_i - \mathbf{c}\|_2 - R)^2.$$

The radial residuals are

$$r_i = \|\mathbf{q}_i - \hat{\mathbf{c}}\|_2 - \hat{R}.$$

The minimum-zone circularity deviation is the smallest radial separation between two concentric circles enclosing all points:

$$t_{\text{CIR}} = \min_{\mathbf{c}} \left[\max_i \|\mathbf{q}_i - \mathbf{c}\|_2 - \min_i \|\mathbf{q}_i - \mathbf{c}\|_2 \right].$$

The analogous minimum-zone sphericity deviation is

$$t_{\text{SPH}} = \min_{\mathbf{c}} \left[\max_i \|\mathbf{q}_i - \mathbf{c}\|_2 - \min_i \|\mathbf{q}_i - \mathbf{c}\|_2 \right],$$

where the optimization is performed in three dimensions. The numerical value depends on the association criterion and any filtering selected in the CMM software; the criterion must therefore be recorded with the result.[3]

2.5 Direction-history and hysteresis indicators

When the same angular position is measured in opposite directions, a matched directional difference can be formed:

$$h(\theta_i) = \varepsilon^+(\theta_i) - \varepsilon^-(\theta_i).$$

Useful summary indicators are

$$H_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{i=1}^N h^2(\theta_i)}, \quad H_{\text{pp}} = \max_i h(\theta_i) - \min_i h(\theta_i),$$

and the mean directional bias

$$\bar{h} = \frac{1}{N} \sum_{i=1}^N h(\theta_i).$$

These quantities require point-to-point angular correspondence. A visual separation between traces is suggestive but is not, by itself, a quantified hysteresis result.

2.6 Thermal sensitivity and uncertainty budget

Length measurements are conventionally referred to 20 °C. For a component with coefficient of thermal expansion α_L ,

$$\Delta L_T = \alpha_L L (T - 20^\circ\text{C}).$$

For illustration, a steel-like coefficient of $\alpha_L \approx 11.5 \mu\text{m}/(\text{m K})$ and $L = 151 \text{ mm}$ give approximately $1.74 \mu\text{m}$ per kelvin. This is of the same order as the reported planar circularity values, so temperature records and artefact equilibration are essential.

A practical uncertainty budget for the present experiment may be written

$$u_c^2 = u_{\text{CMM}}^2 + u_{\text{probe}}^2 + u_{\text{MCG}}^2 + u_T^2 + u_{\text{repeat}}^2 + u_{\text{fit}}^2 + u_{\text{drift}}^2.$$

No numerical expanded uncertainty is claimed in this report because the required calibration certificates, thermal logs, repeatability series, and software-association uncertainty were not available in the original dataset.

Experimental Protocol and Point Generation

WORK PACKAGE 2 Implementation on the ZEISS CMM

3.1 Equipment and available MCG radii

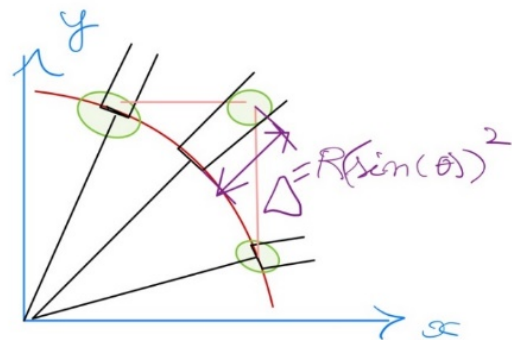
The original point-generation scripts support nominal arm lengths

$$R \in \{101, 151, 226, 380, 532, 685\} \text{ mm.}$$

The first planar campaign uses $R = 151$ mm. The CMM follows a sequence of Cartesian coordinates while maintaining the specified contact normal. The experimental geometry and the original MCG arrangement are shown in figure 3.1.



(a) MCG installed beneath the CMM probe.



(b) Original sketch of the straight-line probe motion between angular stations.

Figure 3.1: Experimental artefact and geometric motivation for trajectory discretization.

Parametric point generation

For a planar verification circle of nominal centre $\mathbf{c} = (x_c, y_c, z_c)^T$ and radius R , the i th nominal point and outward probing normal are generated directly from the angular parameter θ_i :

$$\mathbf{p}_i = \mathbf{c} + R \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \\ 0 \end{bmatrix}, \quad \theta_i = \theta_0 + \frac{2\pi i}{N},$$

$$\mathbf{n}_i = \frac{\mathbf{p}_i - \mathbf{c}}{\|\mathbf{p}_i - \mathbf{c}\|_2}.$$

The implementation validates finite coordinates, unit-normal magnitude, point count, and angular ordering before producing the six-column ZEISS input record

$$[x_i \ y_i \ z_i \ n_{x,i} \ n_{y,i} \ n_{z,i}].$$

The complete Python source is supplied with the project annexes rather than reproduced as screenshots in the report.

3.2 Planar path and probe-retention criterion

Between two consecutive stations separated by $\Delta\theta = 2\pi/N$, the CMM follows the straight chord rather than the circular arc. The commanded travel is therefore

$$d_c = 2R \sin\left(\frac{\Delta\theta}{2}\right) = 2R \sin\left(\frac{\pi}{N}\right).$$

If $d_{\max}$ is the largest admissible displacement that keeps the stylus captured by the MCG fork, the sampling requirement is

$$N \geq \left\lceil \frac{\pi}{\arcsin(d_{\max}/2R)} \right\rceil.$$

The original numerical study obtained $N_{\min} = 30$. The experimental campaign used $N = 60$, reducing the chord length from approximately 31.57 mm to 15.81 mm for $R = 151$ mm and providing a conservative retention margin.

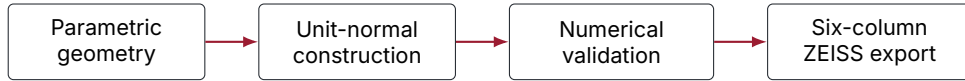


Figure 2: Measurement-point generation and verified ZEISS export workflow.

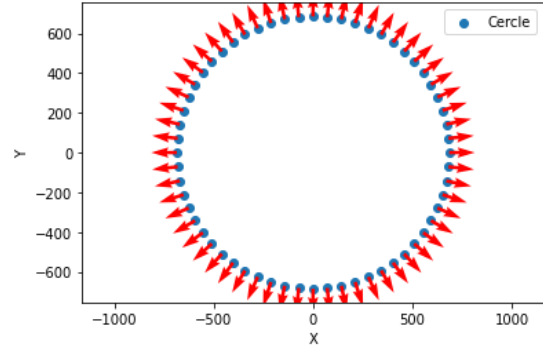


Figure 1: Planar MCG trajectory, radial normals, and probe-retention geometry.

3.3 Spherical-segment path

The spatial trajectory extends the same parametric construction to a set of latitude rings. The code is treated as an implementation asset; the report retains the governing geometry and validation criteria.

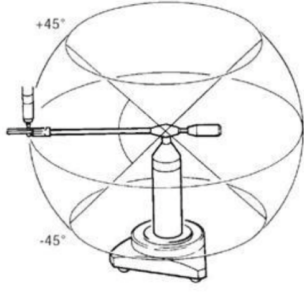


Figure 3: MCG angular envelope and spherical-segment parameterization.

For elevation ϕ_j and azimuth $\theta_{i,j}$, the nominal spherical point is

$$\mathbf{p}_{i,j} = \mathbf{c} + R \begin{bmatrix} \cos \phi_j \cos \theta_{i,j} \\ \cos \phi_j \sin \theta_{i,j} \\ \sin \phi_j \end{bmatrix}, \quad \mathbf{n}_{i,j} = \frac{\mathbf{p}_{i,j} - \mathbf{c}}{R}.$$

The original campaign used ten rings uniformly spaced from -45° to $+45^\circ$,

$$\phi_j = -45^\circ + j \frac{90^\circ}{M-1}, \quad M = 10,$$

with ring radius and height

$$\rho_j = R \cos \phi_j, \quad z_j = R \sin \phi_j.$$

A more uniform chord spacing is obtained by adapting the azimuthal count:

$$N_j = \max \left(N_{\min}, \left\lceil \frac{2\pi\rho_j}{d_t} \right\rceil \right).$$

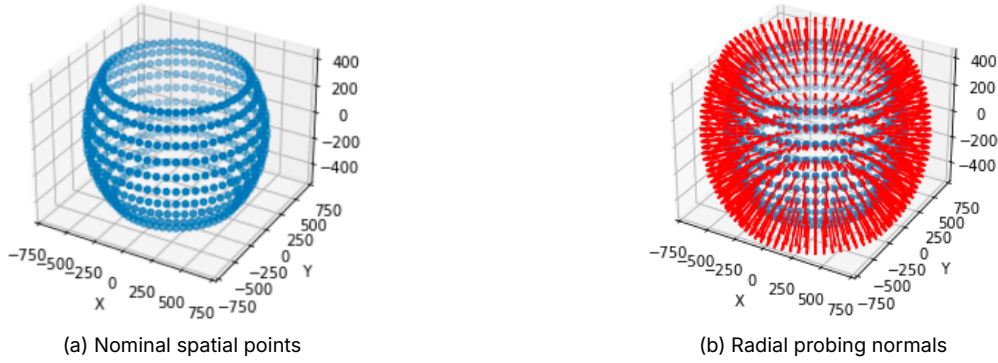


Figure 4: Spatial sampling points and associated radial probing normals.

3.4 ZEISS export and sequence construction

The numerical routine creates one validated file per available arm length. Each row contains the nominal point and a unit approach vector. Before export, the following checks are applied: $\|\mathbf{n}_i\|_2 = 1$ within tolerance, duplicate-point rejection, finite-value verification, preservation of the intended traversal order, and consistent length units. The full Python scripts and representative ASCII files remain available in the annex package.

Implementation principle

The report documents the equations, assumptions, inputs, outputs, and validation rules. Source code is maintained separately so that it can be versioned, tested, and reused without turning the technical narrative into a sequence of editor screenshots.



Figure 5: Six-column measurement schema and sequence-construction workflow.

Field	Meaning	Validation
X, Y, Z	Nominal CMM coordinates of the contact point	finite; common unit
N_X, N_Y, N_Z	Unit probing direction used by the ZEISS routine	$\ \mathbf{n}\ _2 = 1$
Index	Sequence position and implied traversal direction	unique; ordered
Metadata	arm length, speed, probe qualification, timestamp, temperature	recorded per run

3.5 Bidirectional and multi-revolution tests

To expose direction-history effects, the planar path was executed using four deliberately different traversal families:

1. two negative revolutions followed by two positive revolutions, $-720^\circ \rightarrow +720^\circ$;
2. one negative and one positive revolution, $-360^\circ \rightarrow +360^\circ$;
3. two negative revolutions only, $0^\circ \rightarrow -720^\circ$;
4. two positive revolutions only, $0^\circ \rightarrow +720^\circ$.

The spatial sequence combines clockwise and counter-clockwise rings with top-to-bottom and bottom-to-top ordering. This design separates azimuth reversal from vertical traversal reversal, although repeated runs are still required to distinguish hysteresis from time drift.

Recommended data record for every run

Retain the raw deviation file, nominal-point file, path direction, ring order, time stamp, arm length, CMM and artefact temperatures, probe-qualification identifier, stylus configuration, speed and acceleration, fitting criterion, filter state, and operator interventions. Without this metadata, small differences between runs cannot be assigned to a physical mechanism with confidence.

Planar Verification Results

WORK PACKAGE 3 Circularity and direction-history analysis

4.1 Reported form plots

The four original ZEISS plots are reproduced in figure 4.1. The red curve is the associated theoretical circle, while the blue trace represents the magnified measured radial deviation.

4.2 Quantitative comparison

The overall mean of the four reported circularity values is

$$\bar{t}_{\text{CIR}} = 1.2725 \mu\text{m},$$

with a sample standard deviation of approximately 0.189 μm . The reversal-containing paths average

$$\bar{t}_{\text{rev}} = \frac{1.49 + 1.37}{2} = 1.43 \mu\text{m},$$

whereas the unidirectional paths average

$$\bar{t}_{\text{uni}} = \frac{1.10 + 1.13}{2} = 1.115 \mu\text{m}.$$

The unidirectional average is therefore about 22% lower:

$$\frac{\bar{t}_{\text{rev}} - \bar{t}_{\text{uni}}}{\bar{t}_{\text{rev}}} \times 100 \approx 22.0\%.$$

Planar finding

The data support the practical recommendation to use a continuous one-direction path as the baseline verification sequence and to reserve reversal-containing paths for a separate hysteresis diagnostic. The difference is consistent with backlash or mechanical hysteresis, but the present dataset contains

Table 4.1: Reported planar circularity results.

Path	Points	Circularity (μm)	Displayed radius (software value)	Interpretation
$-720^\circ \rightarrow +720^\circ$	240	1.49	1.50073	Two reversals and four total revolutions.
$-360^\circ \rightarrow +360^\circ$	122	1.37	1.50073	One revolution in each direction.
$0^\circ \rightarrow -720^\circ$	120	1.10	1.50073	Continuous negative direction.
$0^\circ \rightarrow +720^\circ$	120	1.13	1.50073	Continuous positive direction.

only one reported result per path family and therefore cannot quantify repeatability or establish statistical significance.

4.3 Metrological cautions

The plot field labelled “tolerance = 0.0000 mm” should not be interpreted as a zero-tolerance acceptance requirement. It indicates that no engineering tolerance was entered in the reporting template. Likewise, the magnification factor of 10,000 affects only the graphical display and not the numerical deviation.

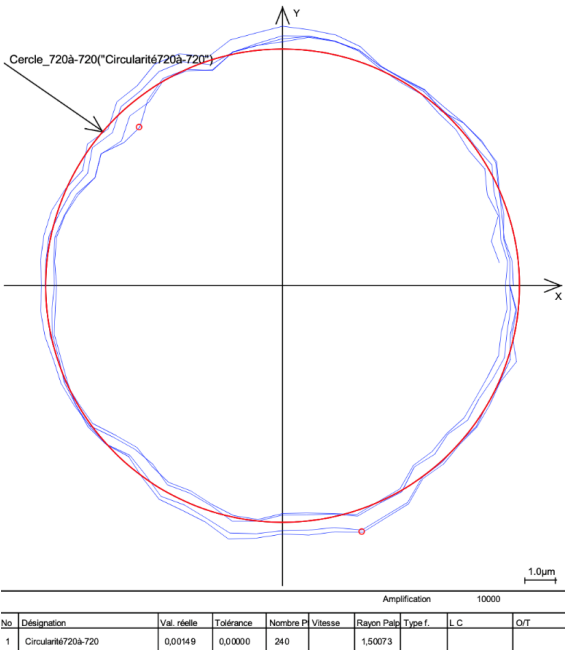
A second inconsistency requires attention. The point-generation section states a nominal arm radius of 151 mm, while the ZEISS result tables are transcribed as a fitted radius of 1.50073 in the displayed software unit. Before the campaign is used for traceable verification, the coordinate units, scale factor, feature-association output, and nominal radius imported into CALYPSO should be audited and recorded.

4.4 Repeatability experiment required for confirmation

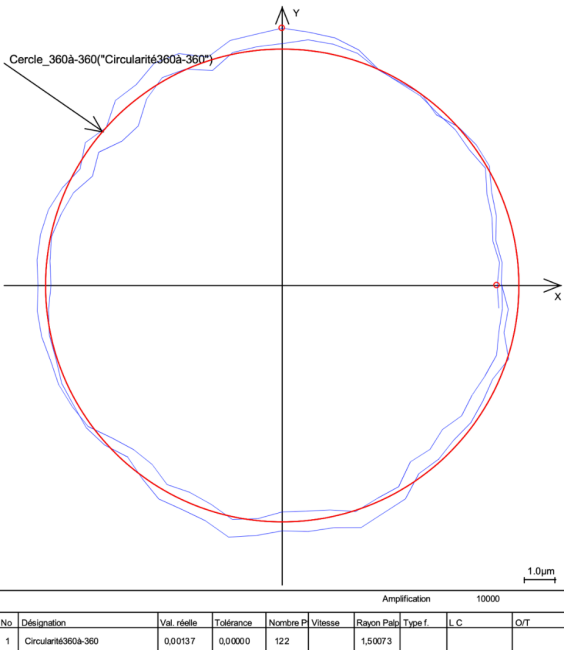
For each path family, perform at least $n = 5$ independent runs after re-approach and, preferably, requalification. If t_j denotes the resulting circularity value,

$$\bar{t} = \frac{1}{n} \sum_{j=1}^n t_j, \quad s_t = \sqrt{\frac{1}{n-1} \sum_{j=1}^n (t_j - \bar{t})^2}.$$

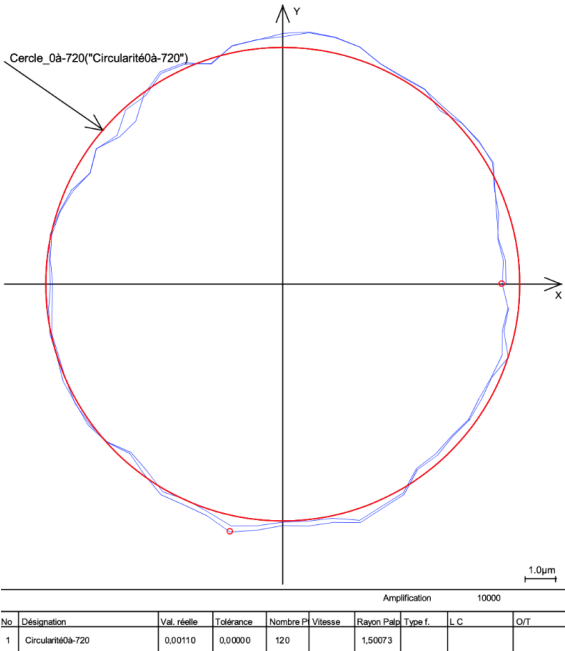
A paired comparison between positive and negative paths should use matched angular residuals rather than only the scalar circularity values. This preserves the spatial structure of the direction-dependent error.



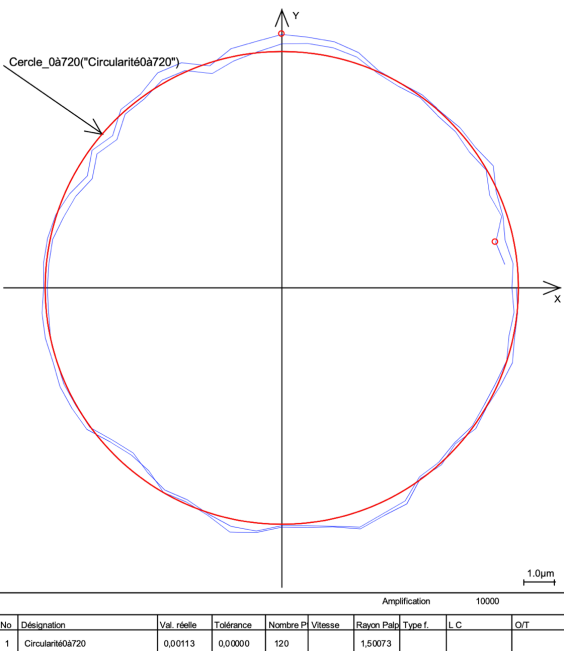
(a) -720° to $+720^{\circ}$.



(b) -360° to $+360^{\circ}$.



(c) 0° to -720° .



(d) 0° to $+720^{\circ}$.

Figure 4.1: Planar MCG circularity results for the four traversal strategies.

Spatial Error Identification and Validation

WORK PACKAGE 4 Least-squares estimation from spherical normal deviations

5.1 Original spherical data model

For an ideal sphere centered at the origin, the CMM returns ε_i along the nominal radial normal. The project assumes that the realized sphere can be represented by a translated center $\mathbf{t} = (u, v, w)^\top$ and an effective radius $R + \delta R$. To first order,

$$\varepsilon_i = un_{x,i} + vn_{y,i} + wn_{z,i} + \delta R + \eta_i,$$

where η_i collects probing noise, form error, unmodeled CMM geometry, and environmental effects.

Defining

$$\mathbf{x} = \begin{bmatrix} u & v & w & \delta R \end{bmatrix}^\top, \quad A_i = \begin{bmatrix} n_{x,i} & n_{y,i} & n_{z,i} & 1 \end{bmatrix},$$

the complete system is

$$\boldsymbol{\varepsilon} = A\mathbf{x} + \boldsymbol{\eta}.$$

The least-squares estimate is

$$\hat{\mathbf{x}} = A^+ \boldsymbol{\varepsilon},$$

where A^+ is the Moore–Penrose pseudoinverse. The normal-equation expression

$$\hat{\mathbf{x}} = (A^\top A)^{-1} A^\top \boldsymbol{\varepsilon}$$

is valid only when A has full column rank and is sufficiently well conditioned. In numerical software, QR or

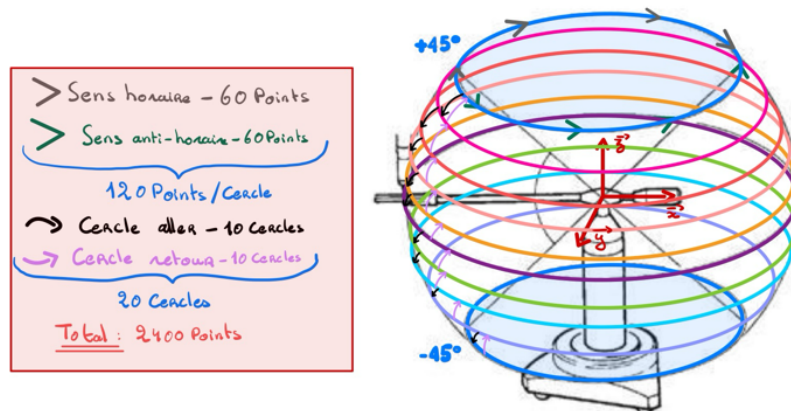


Figure 5.1: Original schematic of the consecutive latitude-circle sequences used in the spatial campaign.

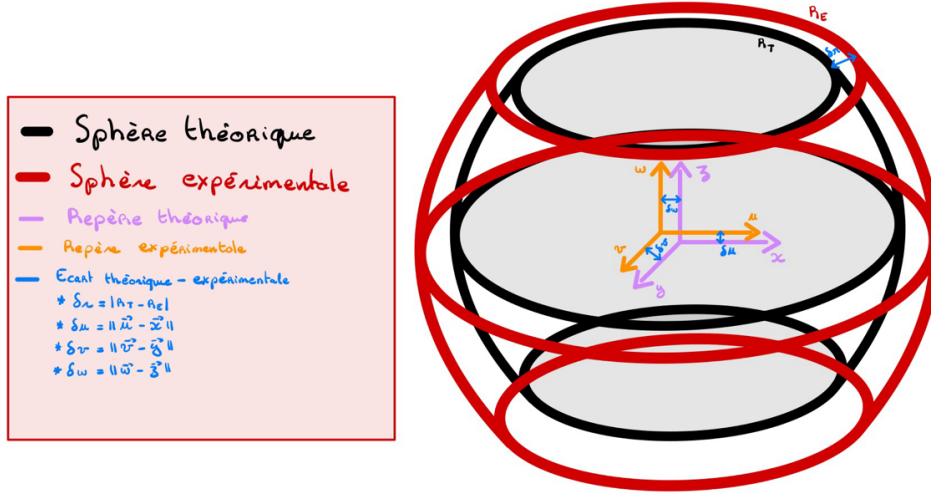


Figure 5.2: Original geometric representation of the ideal sphere and the translated, radius-offset sphere.

singular-value decomposition is preferred because it avoids explicitly forming $(A^T A)^{-1}$.

The residual vector, residual variance, and parameter covariance estimate are

$$\hat{\mathbf{r}} = \boldsymbol{\varepsilon} - A\hat{\mathbf{x}}, \quad \hat{\sigma}^2 = \frac{\|\hat{\mathbf{r}}\|_2^2}{N - p},$$

$$\text{Cov}(\hat{\mathbf{x}}) \approx \hat{\sigma}^2 (A^T A)^{-1}.$$

The condition number $\kappa(A)$ should be reported; a large value indicates that different parameter combinations produce nearly indistinguishable residual fields.

5.2 Small squareness terms

The original project estimates three small angular defects after solving for $u, v, w, \delta R$. Under its sign convention, the predicted residual is extended as

$$\varepsilon_i \approx un_{x,i} + vn_{y,i} + wn_{z,i} + \delta R - \alpha z_i n_{y,i} - \beta x_i n_{z,i} - \gamma y_i n_{x,i} + \eta_i.$$

A joint regression is therefore possible with

$$\mathbf{q} = \begin{bmatrix} u & v & w & \delta R & \alpha & \beta & \gamma \end{bmatrix}^T$$

and row

$$A_i = \begin{bmatrix} n_{x,i} & n_{y,i} & n_{z,i} & 1 & -z_i n_{y,i} & -x_i n_{z,i} & -y_i n_{x,i} \end{bmatrix}.$$

The unified estimate

$$\hat{\mathbf{q}} = A^+ \boldsymbol{\varepsilon}$$

is preferable to solving three independent one-parameter problems because translation, radius, and squareness terms are correlated. The joint model also exposes weak observability directly through the singular values of A .

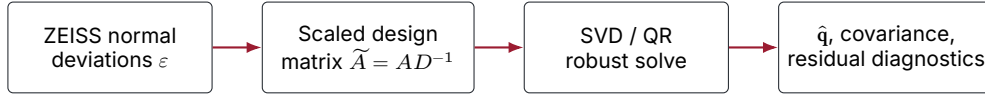


Figure 6: Robust numerical estimation workflow for the seven-parameter model.

5.3 Numerical implementation

The implementation imports the nominal coordinates, unit normals, and scalar ZEISS deviations, constructs the joint seven-parameter model, scales columns to comparable numerical magnitudes, and solves the system without explicitly forming $(A^T A)^{-1}$. Let $D = \text{diag}(\|A_{:,1}\|_2, \dots, \|A_{:,p}\|_2)$ and $\tilde{A} = AD^{-1}$. The scaled estimate is

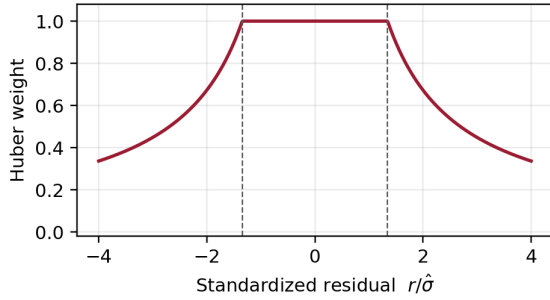
$$\hat{\mathbf{z}} = \tilde{A}^+ \varepsilon, \quad \hat{\mathbf{q}} = D^{-1} \hat{\mathbf{z}}.$$

To reduce the influence of isolated probing anomalies, the solution may be embedded in an iteratively reweighted least-squares loop. For standardized residual $u_i = r_i / \hat{\sigma}$, the Huber weight is

$$w_i = \begin{cases} 1, & |u_i| \leq c, \\ \frac{c}{|u_i|}, & |u_i| > c, \end{cases} \quad c = 1.345,$$

and each iteration solves

$$\hat{\mathbf{q}}^{(k+1)} = \arg \min_{\mathbf{q}} \|W_k^{1/2} (A\mathbf{q} - \varepsilon)\|_2^2.$$



The final report should include the singular values of A , the condition number $\kappa(A)$, residual RMS, leverage or influence diagnostics, and an approximate covariance matrix

$$\text{Cov}(\hat{\mathbf{q}}) \approx \hat{\sigma}^2 (A^T W A)^{-1}.$$

The complete Python implementation and verification scripts are provided in the annex package. The main text therefore focuses on the model, numerical conditioning, and acceptance logic rather than source-code screenshots.

Figure 7: Huber residual weights used by the iteratively reweighted solution.

Model limitation

The seven-parameter model is a first-order diagnostic approximation, not a complete CMM volumetric-error map. Scale, straightness, probe-lobing, elastic, and thermal-gradient terms are omitted. Estimated angular parameters must not be uploaded as compensation values until validated with calibrated artefacts and independent orientations.

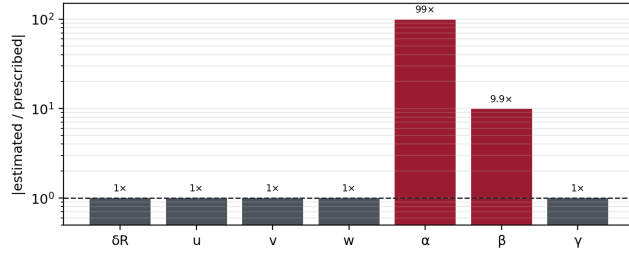


Figure 8: Synthetic parameter-recovery audit of the original identification model.

Parameter	Prescribed	Estimated
δR (mm)	0.100000	0.100000
u (mm)	0.010000	0.009967
v (mm)	0.020000	0.019999
w (mm)	0.030000	0.030000
α (μrad)	0.1	9.879
β (μrad)	5.0	49.387
γ (μrad)	10.0	10.000

Table 1: Synthetic validation values reported in the original project.

5.4 Audit of the synthetic validation

The synthetic test confirms the translation, effective-radius, and γ channels to the reported precision. It does not validate α and β : their recovered magnitudes differ by factors of approximately 99 and 9.9. Such discrepancies cannot be explained by floating-point rounding alone and indicate a unit conversion, axis convention, parameter-coupling, or synthetic-generator inconsistency.

A defensible validation campaign should therefore:

1. generate a full-factorial set of independent translation, radius, and squareness perturbations;
2. solve all parameters jointly using column scaling and SVD;
3. report $\kappa(A)$, singular vectors, parameter covariance, and residual structure;
4. include zero-parameter and single-parameter tests to verify signs and units;
5. validate the recovered correction on an independent synthetic dataset not used for fitting.

$$\mathbf{e}_{\text{val}} = \mathbf{q}_{\text{estimated}} - \mathbf{q}_{\text{prescribed}}, \quad E_j = \frac{|e_{\text{val},j}|}{\max(|q_{\text{prescribed},j}|, q_{j,\min})}.$$

Interpretation

The source code remains valuable as a transparent prototype, but the angular channels must be treated as unvalidated until the scale and convention mismatch is resolved. The appropriate scientific conclusion is not that the model failed globally, but that only a subset of the estimated parameters passed the available verification test.

Table 5.2: Spatial parameter estimates for the four traversal histories. Translational quantities are shown in micrometres and angular quantities in microradians.

Path	δR	u	v	w	α	β	γ
CW, top-down	0.026	0.159	-0.033	-0.087	-0.783	-0.312	0.265
CCW, top-down	-0.001	0.174	0.039	-0.180	-1.400	0.490	1.485
CW, bottom-up	-0.123	-0.133	-0.033	0.359	-0.797	-0.103	2.064
CCW, bottom-up	-0.074	-0.198	0.062	0.179	3.413	-0.601	-1.543

5.5 Reported experimental parameter estimates

The fitted translation and radius terms are sub-micrometric, while the angular terms are of microradian order. However, the sign and magnitude change substantially with both azimuth direction and vertical ring order. This path dependence is consistent with mechanical hysteresis, but it may also contain thermal drift, probe re-approach effects, qualification error, coordinate-frame mismatch, and model inadequacy. The results should therefore be reported as direction-specific diagnostic estimates rather than as fixed machine constants.

Spatial finding

The experiment demonstrates that large-radius spherical paths are sensitive to very small coordinate-system changes, but the current angular identification is not yet sufficiently validated for compensation. The immediate value of the method is comparative: tracking whether the same path-specific signatures remain stable over time.

Spatial Coverage and Verification Strategy

WORK PACKAGE 5 Improved protocol for periodic CMM monitoring

6.1 Coverage of the $\pm 45^\circ$ spherical band

The area of a spherical zone between elevations ϕ_1 and ϕ_2 is

$$A_{\text{zone}} = 2\pi R^2 (\sin \phi_2 - \sin \phi_1).$$

For $\phi_1 = -45^\circ$ and $\phi_2 = 45^\circ$,

$$A_{\pm 45} = 4\pi R^2 \sin 45^\circ = 2\sqrt{2}\pi R^2.$$

Relative to a complete sphere,

$$\frac{A_{\pm 45}}{4\pi R^2} = \sin 45^\circ \approx 70.7\%.$$

The two omitted polar caps each represent approximately 14.6% of the sphere. Repositioning the MCG to generate complementary upper and lower spherical segments can therefore extend the sampled region without requiring a single mechanically inaccessible full-sphere path.

6.2 Equal-area ring placement

Uniform elevation increments concentrate points near the poles because the ring circumference decreases with $|\phi|$. A simple equal-area strategy samples uniformly in z :

$$z_j = z_{\min} + j \frac{z_{\max} - z_{\min}}{M - 1}, \quad \rho_j = \sqrt{R^2 - z_j^2}.$$

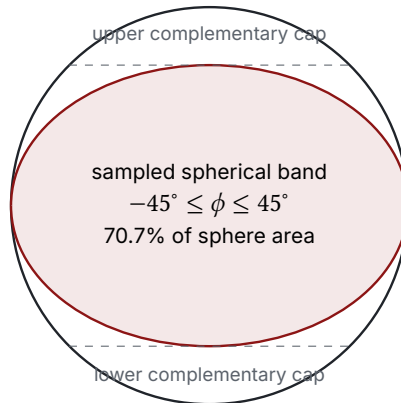


Figure 6.1: Geometric interpretation of the original $\pm 45^\circ$ spherical segment and the complementary-cap strategy.

The number of azimuthal points on ring j is selected from the target chord spacing d_t :

$$N_j = \max\left(N_{\min}, \left\lceil \frac{2\pi\rho_j}{d_t} \right\rceil\right).$$

This produces a more nearly uniform area per point, improves the conditioning of the least-squares model, and avoids unnecessary measurement time on short rings.

6.3 Direction-separated verification design

A periodic campaign should distinguish baseline geometry, direction-history effects, and time drift. A balanced ABBA order is recommended:

$$A_1 \rightarrow B_1 \rightarrow B_2 \rightarrow A_2,$$

where A and B are opposite traversal directions. Matched residuals are interpolated to a common angular grid and summarized by mean bias, repeatability standard deviation, H_{RMS} , and H_{pp} .

6.4 Diagnostic filtering method

Filtering is applied only to a copy of the residual data for visualization and frequency-content interpretation. Conformity and form-error decisions remain based on the unfiltered data. The recommended diagnostic pipeline has two stages.

Stage 1 - Hampel outlier screening

For a local window $\mathcal{W}_k$, define the median and robust scale

$$m_k = \text{median}_{j \in \mathcal{W}_k}(e_j), \quad s_k = 1.4826 \text{ median}_{j \in \mathcal{W}_k} |e_j - m_k|.$$

An isolated sample is replaced by m_k only when

$$|e_k - m_k| > 3s_k.$$

This rejects single-point spikes without smoothing the complete trace.

Stage 2 - Savitzky-Golay shape-preserving smoothing

A local polynomial of degree $d = 3$ is fitted over a symmetric window of $2M + 1 = 21$ points:

$$\hat{\mathbf{a}}_k = \arg \min_{\mathbf{a}} \sum_{j=-M}^M \left[e_{k+j} - \sum_{p=0}^d a_p j^p \right]^2, \quad e_f[k] = \hat{a}_{0,k}.$$

Unlike a simple moving average, this method preserves local slopes and curvature more effectively. The angular sampling interval, window length, polynomial degree, boundary policy, and outlier threshold must be stored with every filtered result.

Filtering and conformity

The filter output is a diagnostic view, not a corrected measurement result. Raw deviations, raw circularity or sphericity, and the exact filter configuration must always be retained.

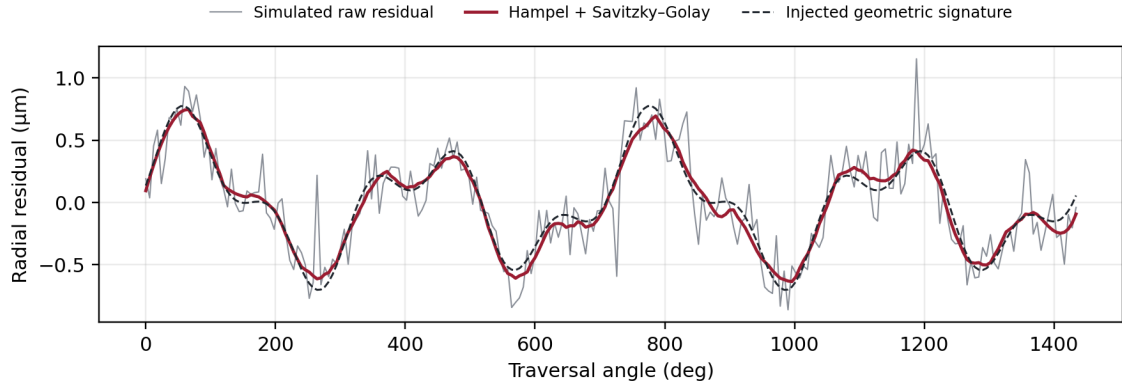


Figure 9: Simulated diagnostic filtering of a planar residual sequence. The injected geometric signature is shown only to quantify filter fidelity and would not be known in a real campaign.

Diagnostic quantity	Raw	Filtered	Change	Interpretation
RMSE to injected signature (μm)	0.187	0.058	−68.9%	random/outlier component reduced
Peak-to-valley residual (μm)	2.017	1.383	−31.4%	isolated peaks suppressed
Correlation with signature	–	0.988	–	repeatable lobing preserved

The simulation uses 240 points over four revolutions, Gaussian noise with $0.18\ \mu\text{m}$ standard deviation, three isolated spikes, and a multi-lobe geometric component. The Hampel stage removes the spikes, while the cubic Savitzky–Golay stage recovers the low-frequency form signature with a correlation of 0.988. The result supports the method for diagnostic visualization. It does *not* justify replacing the raw circularity result by the filtered peak-to-valley value.

6.5 Proposed periodic verification protocol

Phase 1 - Preparation and preconditioning

1. Verify calibration status of the CMM, MCG, probe, and temperature sensors.
2. Stabilize the machine and artefact near 20°C ; record thermal conditions.
3. Qualify the stylus and execute a standardized warm-up trajectory.

Phase 2 - Baseline planar campaign

1. Run at least five positive and five negative two-revolution paths.
2. Fix speed, acceleration, point density, and approach policy.
3. Report raw circularity, residual RMS, repeatability, and matched direction differences.

Phase 3 - Spatial campaign

1. Use equal-area rings over $-45^\circ \leq \phi \leq 45^\circ$.
2. Run CW/CCW and top-down/bottom-up paths in an ABBA schedule.
3. Solve the joint model by scaled QR/SVD and report $\kappa(A)$, uncertainty, and residual diagnostics.
4. Repeat with complementary upper and lower MCG positions.

Phase 4 - Decision and trending

Establish warning and action limits from repeated baseline campaigns. Investigate any result exceeding a control limit, changing the direction-history metric materially, or producing an uncertainty interval incompatible with the decision rule.

6.6 Expected benefits and limitations

The proposed method offers improved spatial coverage, better separation of repeatability and reversal effects, and a mathematically transparent parameter fit. Its principal limitations are the simplified geometric error model, the absence of a calibrated full-volume reference, and the time required for repeated spatial paths. These limitations are manageable when the MCG is used as a stable periodic-monitoring artefact and the results are trended rather than treated as an absolute volumetric calibration.

Conclusion and Recommendations

This project demonstrates a complete digital-to-physical workflow for large-radius CMM verification: nominal circle and sphere points are generated in Python, paired with unit normals, exported to the ZEISS environment, executed along multiple traversal histories, and analyzed through form plots and least-squares identification.

The planar study reports circularity deviations between 1.10 μm and 1.49 μm . The lower values obtained on continuous one-direction paths support the operational recommendation to avoid direction reversals in the baseline sequence and to evaluate reversals separately as a hysteresis test. Nevertheless, additional repeated runs and matched-angle analysis are required before the observed difference can be attributed uniquely to mechanical hysteresis.

The spatial study provides a useful first-order model for center offset, effective radius, and axis squareness. Translation and radius recovery are convincing in the synthetic test, but the α and β validation channels contain order-of-magnitude inconsistencies. Consequently, the present angular estimates should be treated as diagnostic only. A joint SVD solution, explicit axis convention, unit tests, uncertainty propagation, and conditioning analysis are required before compensation is considered.

The most significant methodological improvement is the proposed coverage protocol: equal-area spherical rings, complementary upper and lower segments, unidirectional baseline paths, ABBA reversal tests, repeated runs, thermal monitoring, raw-data retention, and a GUM-consistent uncertainty budget. Implemented together, these measures transform the MCG from a one-off experimental artefact into a defensible periodic-monitoring system for detecting changes in CMM behavior over time.

Priority actions
1. Resolve the nominal-radius and ZEISS display-unit inconsistency. 2. Correct and revalidate the α and β synthetic identification channels. 3. Replace explicit matrix inversion with QR/SVD and report the condition number. 4. Establish repeated-run control limits for the unidirectional baseline path. 5. Add temperature, probe qualification, speed, fit criterion, and filter settings to every data record. 6. Use complementary spherical segments to extend coverage while maintaining accessible probe geometry.

Original Software and Data Assets

The submitted project identifies the following files:

- Python: `generation_points_theorique.py`, `resolution_numerique.py`, and `coordonnees_theorique.py`;
- MATLAB: `Lissage.m`;
- ASCII data: `coordonnees_theorique.txt`, `coordonnees_test.txt`, and `SphereAllerRetour_10042024.txt`.

The screenshots included in the main text preserve the original implementation. For reproducible future campaigns, the scripts should additionally export a machine-readable metadata file containing the configuration and environmental fields listed in [chapter 3](#).

Recommended Output Schema

A future result file should contain at least:

Field	Description
Run identifier	Unique campaign and sequence identifier.
Timestamp	UTC or local time with time zone.
Artefact configuration	Arm length, mounting orientation, and MCG serial number.
CMM configuration	Machine identifier, software version, probe head, stylus, and qualification ID.
Trajectory	Ring index, azimuth, elevation, path direction, and vertical order.
Nominal data	x, y, z, n_x, n_y, n_z .
Measured data	Scalar normal deviation ε and reconstructed point $\mathbf{q}$.
Environment	Machine, artefact, and room temperatures; humidity if monitored.
Processing	Fit criterion, outlier policy, filter definition, and software revision.
Uncertainty	Component standard uncertainties, combined standard uncertainty, and coverage factor.

References

- [1] International Organization for Standardization, *ISO 10360-2:2009, Geometrical Product Specifications (GPS) - Acceptance and Reverification Tests for Coordinate Measuring Machines (CMM) - Part 2: CMMs Used for Measuring Linear Dimensions*, ISO, Geneva, 2009.
- [2] International Organization for Standardization, *ISO 10360-5:2020, Geometrical Product Specifications (GPS) - Acceptance and Reverification Tests for Coordinate Measuring Systems (CMS) - Part 5: CMMs Using Single and Multiple Stylus Contacting Probing Systems*, ISO, Geneva, 2020.
- [3] International Organization for Standardization, *ISO 1101:2017, Geometrical Product Specifications (GPS) - Geometrical Tolerancing - Tolerances of Form, Orientation, Location and Run-out*, ISO, Geneva, 2017.
- [4] International Organization for Standardization, *ISO 14253-1:2017, Geometrical Product Specifications (GPS) - Inspection by Measurement of Workpieces and Measuring Equipment - Part 1: Decision Rules for Verifying Conformity or Nonconformity with Specifications*, ISO, Geneva, 2017.
- [5] Joint Committee for Guides in Metrology, *JCGM 100:2008, Evaluation of Measurement Data - Guide to the Expression of Uncertainty in Measurement*, BIPM, Sèvres, 2008.
- [6] R. J. Hocken and P. H. Pereira, eds., *Coordinate Measuring Machines and Systems*, 2nd ed., CRC Press, Boca Raton, 2012.
- [7] J. A. Bosch, *Coordinate Measuring Machines and Systems*, Marcel Dekker, New York, 1995.
- [8] Renishaw plc, *Machine Checking Gauge System Documentation and Operating Guidance*, Renishaw, Gloucestershire, UK.
- [9] Carl Zeiss Industrial Metrology, *ZEISS CALYPSO Operating and Evaluation Documentation*, ZEISS Industrial Quality Solutions.